

Doom PWAD QA Report

Run 71f75ea6-4f08-4e35-9f1d-e4fe103b13bd | Map MAP01 | Outcome stuck

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness. The test was designed to evaluate navigation, combat engagement, and resource balance, resulting in a 'stuck' outcome after 224 seconds of runtime.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'stuck' after visiting 10 coarse position cluster(s).
- 0 kill(s) recorded against 12 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0833.
- 0 secret(s) found.

The test run ended with an 'stuck' outcome. There were zero fallback actions (total_actions_taken 32 vs total_llm_calls 32). No MCP errors were reported, but the automated playthrough experienced recurring navigation stalls preventing map completion.

Test Item And Environment

This test covered MAP01 in the Freedoom2 IWAD on difficulty level 1. The map contains 12 enemies (Imps, Demons), 12 items, and 4 secret sectors, with dimensions of 3840x3840 units. The map presents a high-complexity static layout with significant resource-to-enemy imbalances.

The test environment utilized ViZDoom for engine emulation, with the Gemini-3.1-flash-lite model managing the automated playthrough via the MCP tool interface. The run was conducted at difficulty level 1 with a 500-tick maximum duration cap.

Metric	Value	Metric	Value
Status	completed	Duration	224
Actions	32	LLM calls	32
Final HP	100	Kills	0
Position samples	127	Movement units	989.5
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/71f75ea6-4f08-4e35-9f1d-e4fe103b13bd.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	12	12	0 hidden by skill/multiplayer flags
Items	12	12	0 hidden by skill/multiplayer flags
Secrets	4	4	Sector special 9
Geometry	208 linedefs	30 sectors	3840 x 3840 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0833.

Coverage Evaluation

Coverage was limited to the starting area and immediate vicinity. No secrets were found, and the automated playthrough did not engage or defeat any of the 12 spawned enemies. The testing reach was insufficient to provide a comprehensive evaluation of the later map stages.

Objectives Covered

- **Initial resource collection:**

Objectives Omitted

- **Map navigation:** Playthrough stalled at tick 179 due to repeated stuck events
- **Combat engagement:** 0 kills despite 12 spawned enemies
- **Secret discovery:** 0 secrets found despite 4 secret sectors in static analysis

Uncovered attributes: The playthrough could not test mid-to-late map geometry, door functionality, or teleporters due to the navigation stall in the starting region.

Defects And Risks

A total of 2 defects were detected, consisting of 1 critical and 1 minor issue. The most impactful defect was a navigation softlock that prevented the automated playthrough from progressing beyond the initial area.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
1	1	softlock_navigation	71	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an automated control loop that should be reviewed with the recording and action trace.
3	3	unreachable_secret	179	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Starting area (-1065, 297):** high The automated playthrough encountered a recurring softlock at this coordinate, preventing further exploration.

Activity Log Summary

1) Static analysis phase (Pre-run), 2) Map exploration phase (ticks 38-179), 3) Defect detection phase (ticks 71, 179), 4) Report generation phase (post-run).

Exploration was severely curtailed by navigation failures, resulting in the automated playthrough repeatedly attempting to move through impassable geometry.

Tick	Event	Position	HP	MCP Action	Stop reason
71	stuck	-1065.3, 297.4	100	move_to	arrived
73	stuck	-1060.0, 292.1	100	take_action	tics_complete
111	item_pickup	-1078.0, 33.6	100	move_to	arrived
119	stuck	-1078.0, 33.6	100	explore	item_found
121	stuck	-1078.0, 33.6	100	explore	item_found
123	stuck	-1078.0, 33.6	100	explore	item_found
167	stuck	-739.9, -103.0	100	aim_and_shoot	out_of_ammo
169	stuck	-732.6, -105.0	100	take_action	tics_complete
179	stuck	-732.6, -105.0	100	explore	enemy_spotted

#	Ticks	Tool	Stop reason	Reasoning summary
0	14 -> 38	move_to	arrived	Initial exploration of the starting area to secure nearby resources before engaging visible threats.
1	39 -> 40	move_to	arrived	Starting area contains easily accessible ammo; securing the nearby CellPack to bolster resource reserves before moving into potential combat zones.
2	41 -> 42	move_to	arrived	Secured the nearby CellPack to improve ammo logistics. Now moving to explore the surrounding area to identify potential threats and progress deeper into the map.
3	43 -> 44	explore	item_found	Secured the initial CellPack resource. Initiating exploration towards the center to locate enemies and map out paths as indicated by navigation_info.
4	45 -> 61	move_to	arrived	Securing local resource pickups before engaging the distant clusters of monsters; moving to collect the visible Chainsaw to enhance close-range capabilities.
5	62 -> 63	move_to	arrived	Collecting the nearby Chainsaw to bolster melee self-defense before engaging the monster clusters.
6	64 -> 65	move_to	arrived	The player is currently in a safe zone with visible weapons. Moving to acquire the Chainsaw (id 8) to secure resources before venturing into the main area where monsters are located.
7	66 -> 67	move_to	arrived	Collecting the remaining accessible weapon resource (Chainsaw id 8) to optimize inventory before approaching the monster clusters in the main map area.
8	68 -> 69	move_to	arrived	Collecting the nearest Chainsaw resource to consolidate starting equipment before scouting the main map area.

9	70 -> 71	move_to	arrived	I am picking up the nearby Chainsaw (id 8) to secure initial resources before moving to explore the main map area to engage the detected demons and imps.
10	72 -> 73	take_action	tics_complete	Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. I am facing a fresh unexplored direction first, then I will move in short bounded steps.
11	74 -> 87	move_to	arrived	I am acquiring the Infrared item (id 9) to improve visibility before navigating toward the combat zone, as initial pathing resulted in a minor movement stall.
12	88 -> 89	explore	item_found	I have acquired the Infrared item as planned. With no starting ammunition, I will now explore to locate weapons or critical pickups, avoiding the nearby demons until I am armed.
13	90 -> 111	move_to	arrived	I have collected the Infrared item. Since I currently lack projectile ammunition, I will prioritize navigation to locate the Chainsaw (id 24) or other weapons to handle the nearby Demon threats effectively.
14	112 -> 113	move_to	arrived	I have no ammunition and am currently unarmed. I am moving to pick up the Chainsaw (id 24) to establish a means of defense before engaging the visible enemy threat.
15	114 -> 115	move_to	arrived	I have successfully navigated to the immediate vicinity of the Chainsaw pickup to resolve the current ammunition-deprived state. I will now pick up the weapon to prepare for the nearby Demon threat.

16	116 -> 117	explore	item_found	I have collected the Chainsaw (id 24) to secure a weapon. Now that I am armed, I will explore the area to find combat engagements.
17	118 -> 119	explore	item_found	I have acquired the chainsaw and am currently unthreatened by nearby enemies. I will now explore to locate additional ammunition and engage the Imp identified at id 27.
18	120 -> 121	explore	item_found	I have acquired the Chainsaw for close-quarters defense. I am now proceeding to explore the map toward the east to find ammunition and engage the identified enemies.
19	122 -> 123	explore	item_found	The previous recovery did not create measurable progress, so I am exploring from the new angle with enemy and item stops enabled.
20	124 -> 157	move_to	enemy_nearby	I am currently at 0 ammo for my ranged weapons and have only a chainsaw. I am moving toward the nearby ClipBox to secure ranged capabilities before attempting to engage the identified Imp and Demon threats.
21	158 -> 159	move_to	enemy_nearby	I am moving to collect the visible ClipBox to resolve the current lack of ranged ammunition before engaging the nearby DoomImp.
22	160 -> 161	move_to	enemy_nearby	I am moving to the ClipBox to secure ammunition, as currently, I only have a chainsaw and cannot effectively engage the identified threats.
23	162 -> 163	move_to	enemy_nearby	I am collecting the ClipBox to address the ammunition starvation, which is necessary to engage the visible Imp and clear the path forward.

24	164 -> 165	aim_and_shoot	out_of_ammo	I have successfully collected the ClipBox to restore ranged capabilities. I will now target the visible DoomImp (id: 27) to begin clearing threats.
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Limitations And Recommendations

The automated playthrough did not reach any locked doors, teleporters, or crusher traps.

Good Quality Areas

- **Resource initialization:** The automated playthrough correctly identified and prioritized the collection of available weapons like the Chainsaw in the starting area.